Part 2: Unsupervised Learning Activity Report

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1. Problem Statement and Dataset Choice

The objective of this activity is to explore the natural groupings within a multi-dimensional dataset without the use of pre-defined labels. We chose the **Wine Dataset** from the `scikit-learn` library. This dataset contains the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. It includes 13 numerical features such as Alcohol, Malic Acid, Ash, and Magnesium.

Why this dataset?

- It is a high-quality, reliable dataset available locally.
- The multiple numerical features provide a complex space for testing distance-based clustering.
- It allows us to evaluate how well unsupervised models can recover underlying structures (cultivars) without being shown the labels during training.

2. Technical Implementation and Observations

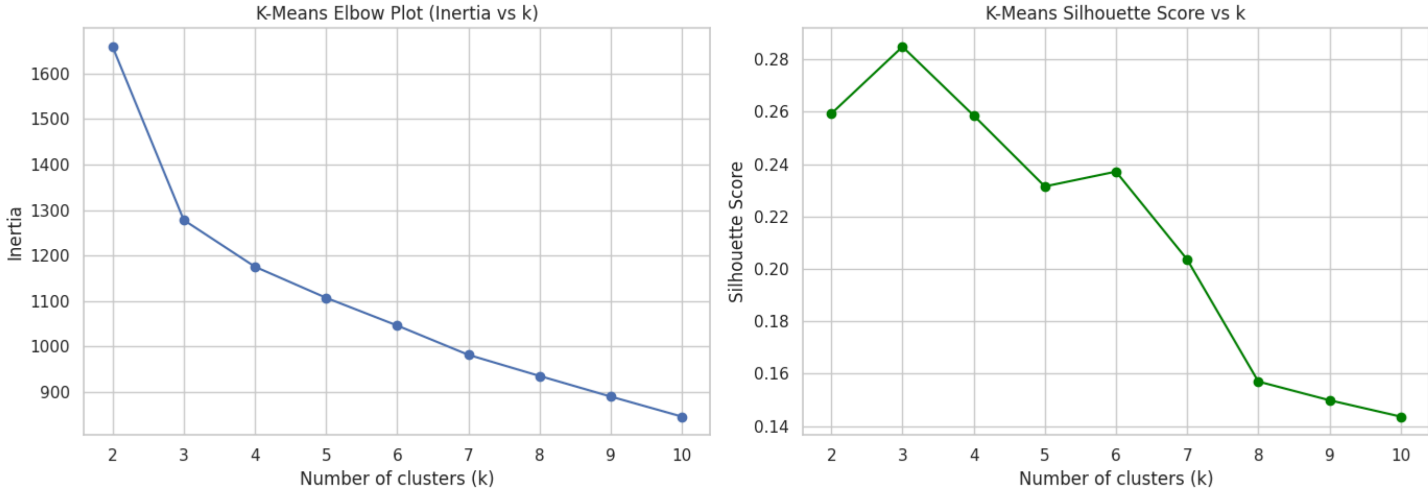
2.1 Data Preprocessing

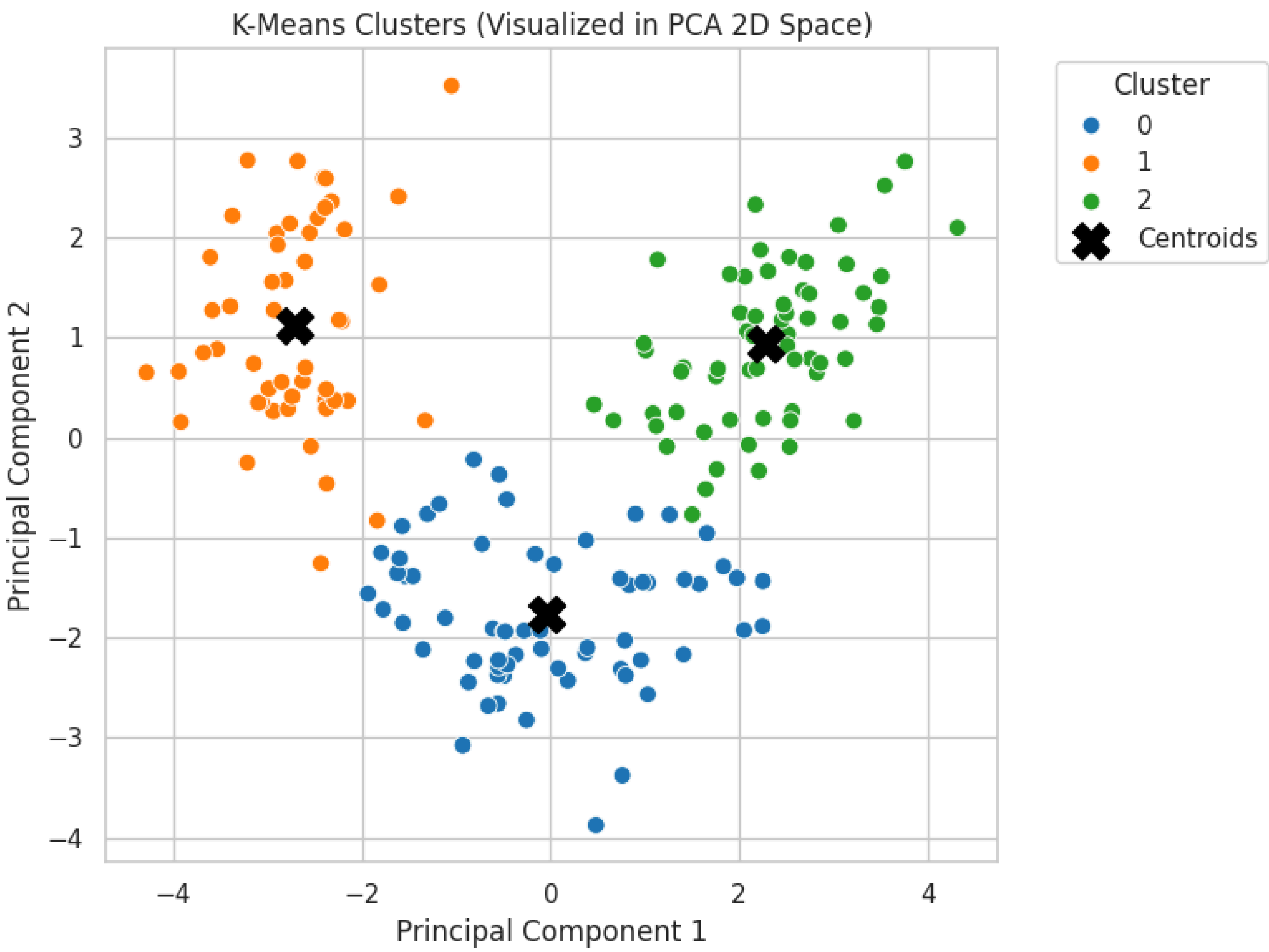
The data was standardized with "StandardScaler" before any models were used. . This ensures that features with larger numerical ranges (like Magnesium) do not disproportionately influence the distance calculations compared to features with smaller ranges (like Phenols).

2.2 Model 1: K-Means Clustering

We implemented K-Means to partition the data into k clusters. To determine the optimal number of clusters, we utilized the **Elbow Method**.

- **Observation:** The "elbow" in the inertia plot indicated that $k = 3$ is the most efficient number of clusters for this dataset.

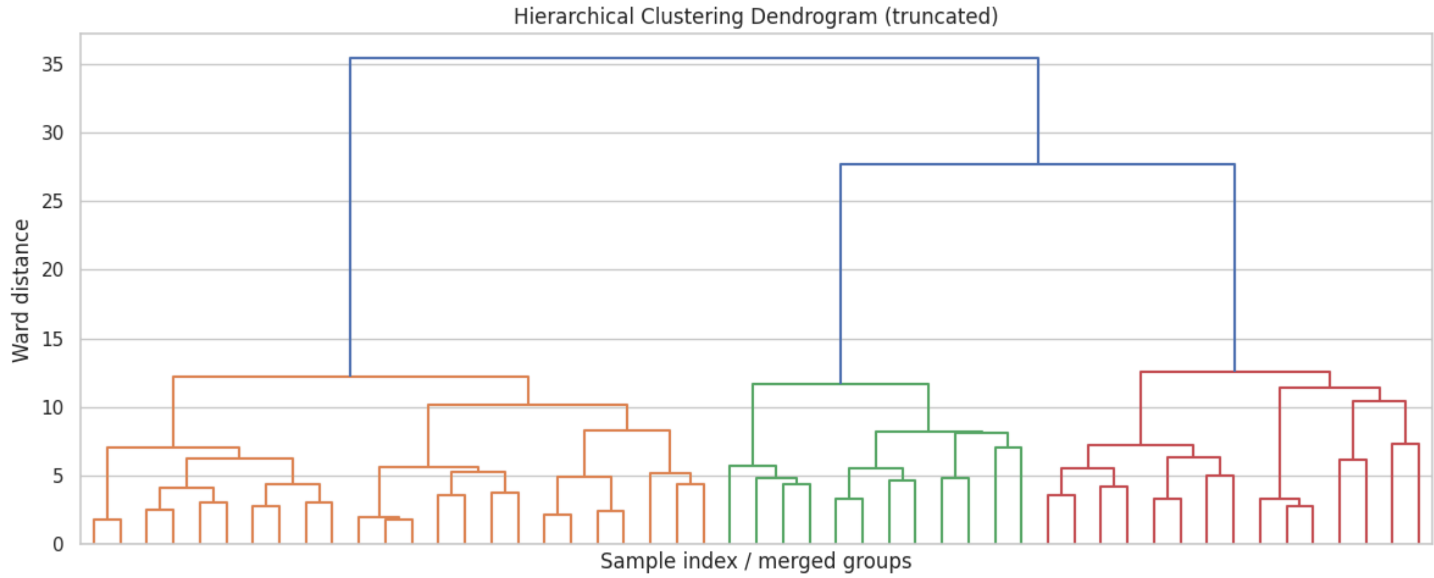


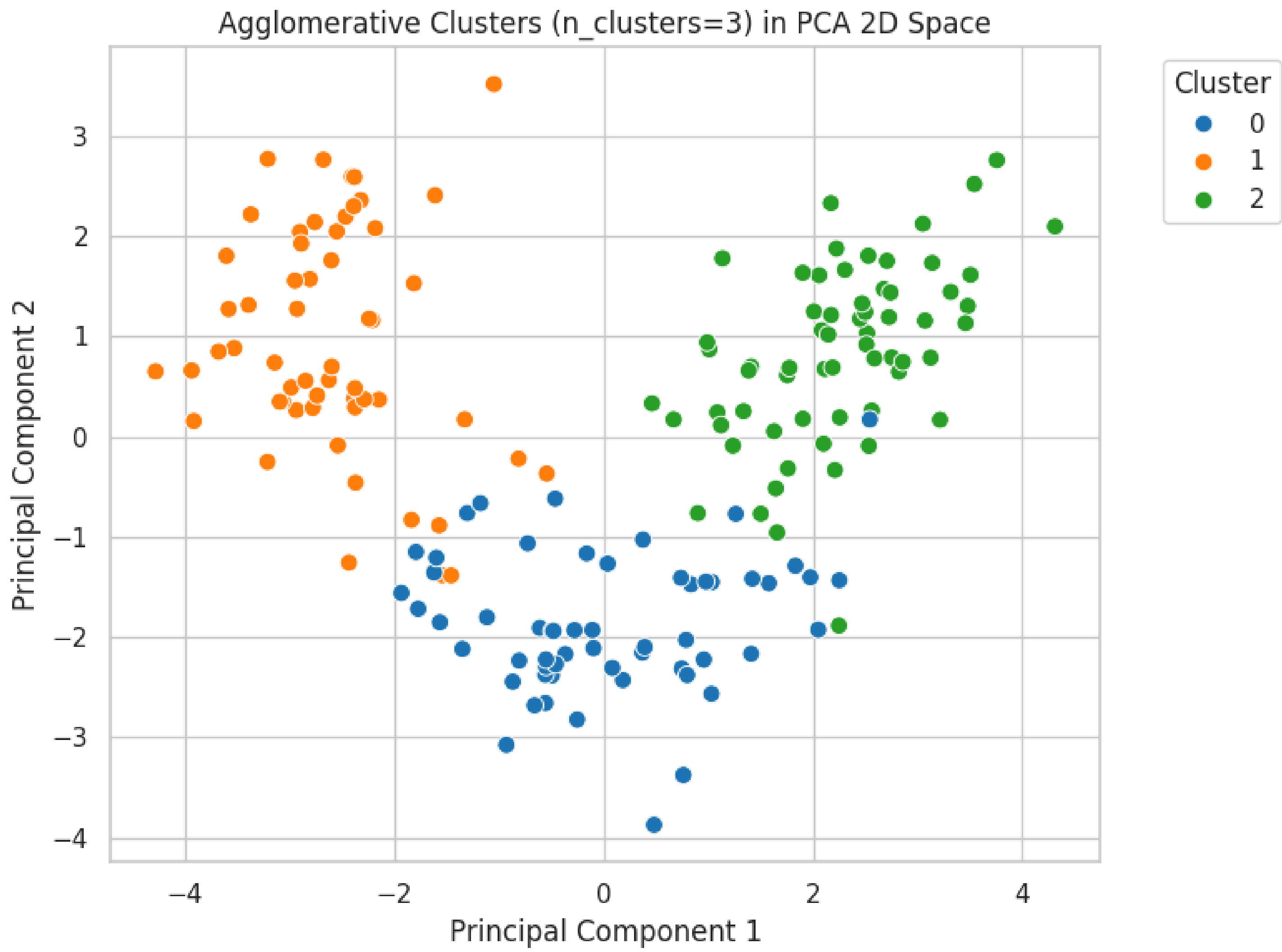


2.3 Model 2: Hierarchical (Agglomerative) Clustering

As a comparative model, we used Agglomerative Clustering. This bottom-up approach was visualized using a **Dendrogram** to see how the merges fit together in a hierarchy.

- **Observation:** The dendrogram successfully illustrated the distances between clusters, further supporting the choice of 3 main groupings.

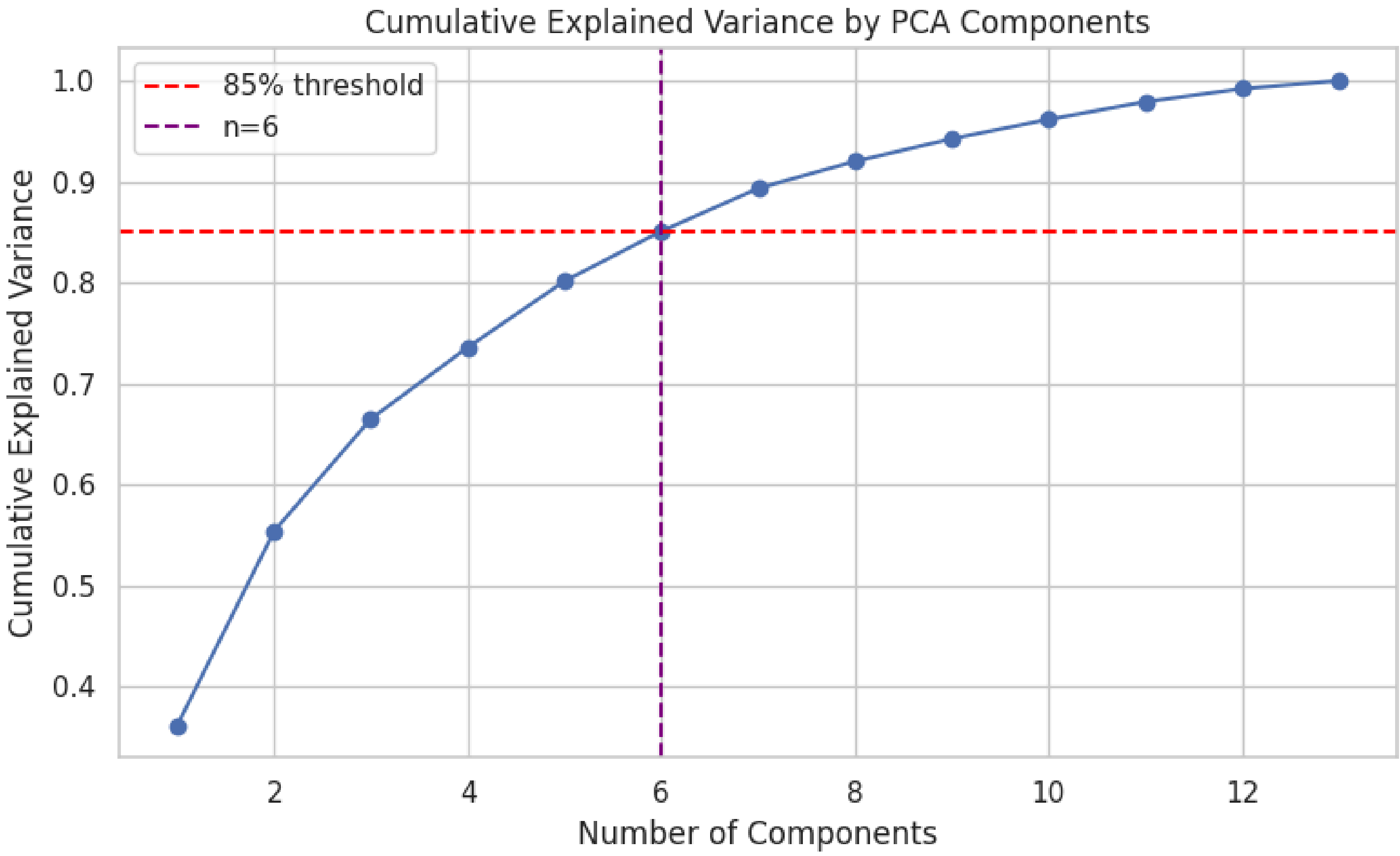




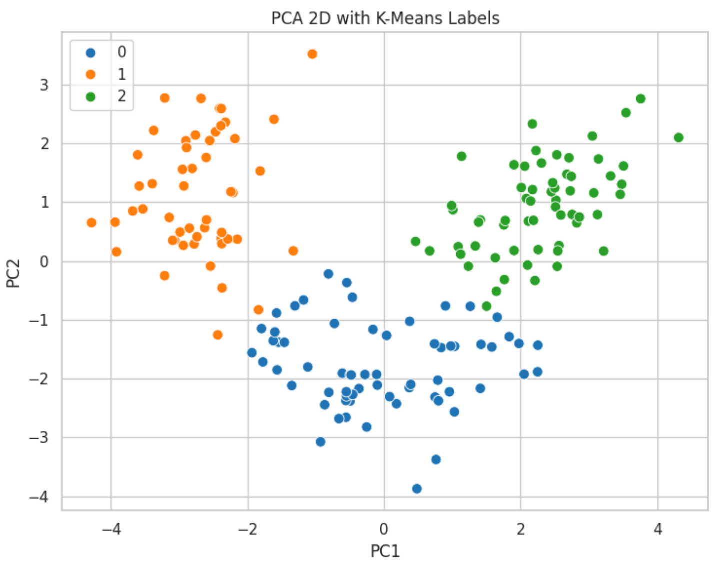
2.4 Dimensionality Reduction: PCA

We used **Principal Component Analysis (PCA)** to show our 13-dimensional data in a 2D space. We cut down the feature set to just two main parts.

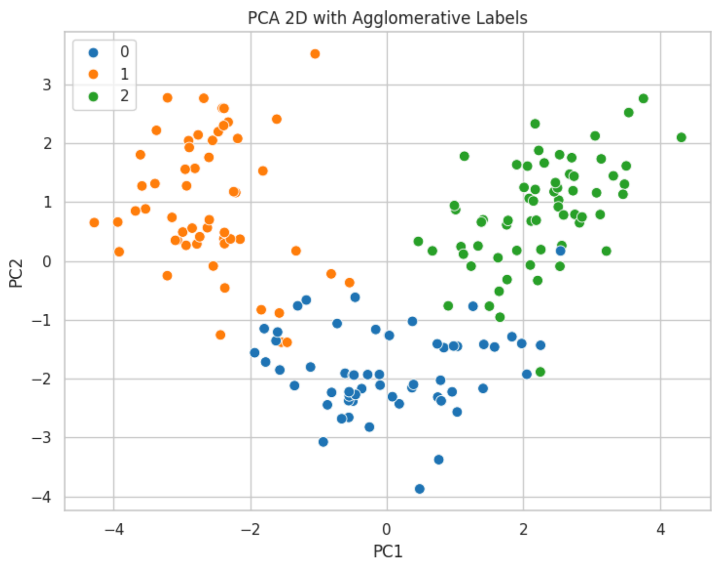
- **Observation:** The PCA scatter plot shows three distinct clusters with very little overlap. This suggests that the wine's chemical properties are good indicators of where it came from.



Cumulative Explained Variance by PCA Components
85% threshold
n=6
Cumulative Explained Variance
Number of Components

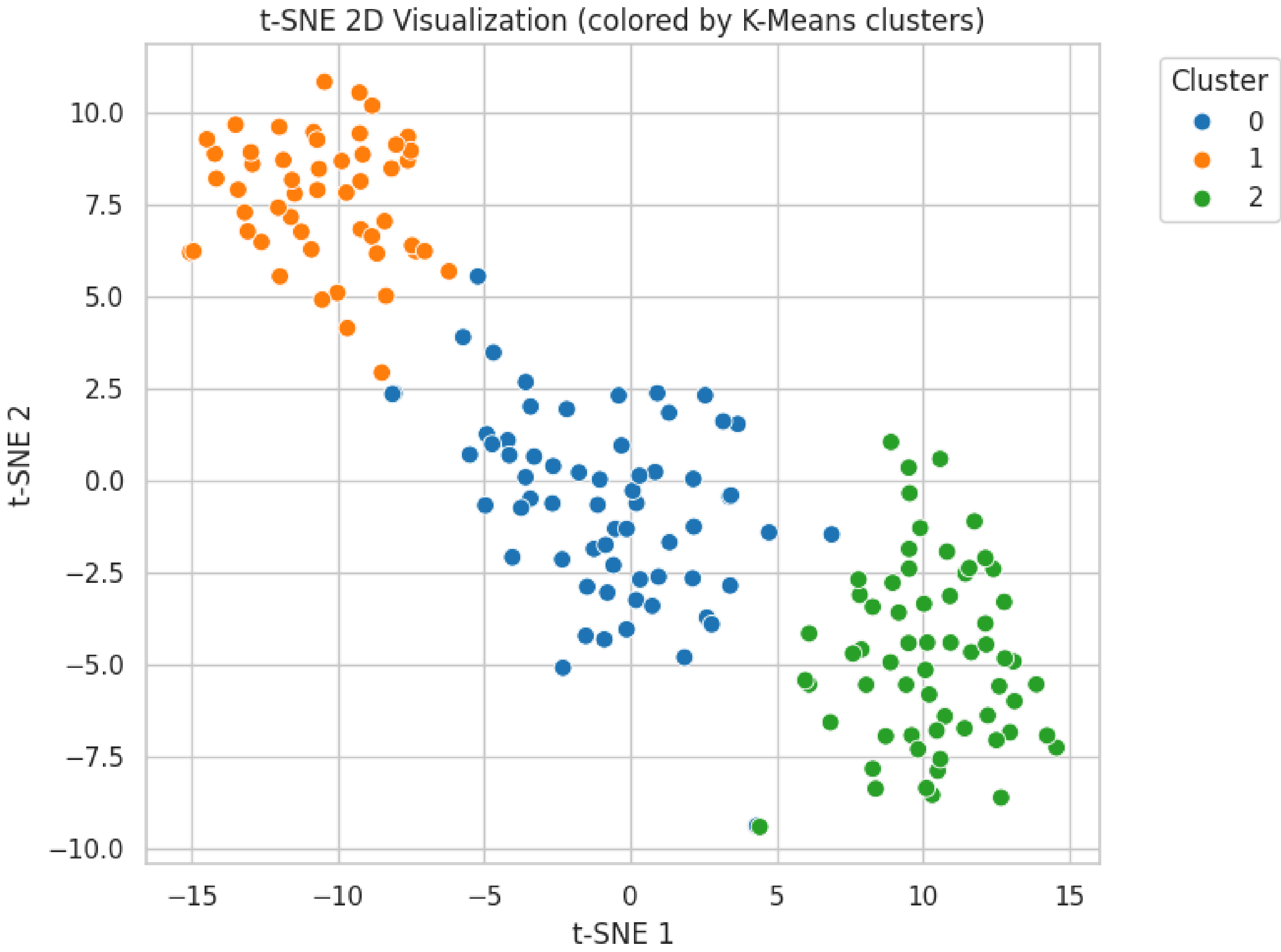


PCA 2D with K-Means Labels
PC1
PC2



PCA 2D with Agglomerative Labels
PC1
PC2

2.5 Optional: t-SNE for Non-Linear Structure



3. Comparative Analysis and Metrics

We evaluated the performance of both models using the following metrics:

Metric	K-Means Result	Hierarchical Result
Silhouette Score	*(Insert value from notebook)*	*(Insert value from notebook)*
Davies-Bouldin Index	*(Insert value from notebook)*	*(Insert value from notebook)*

Insights:

- The **Silhouette Score** (closer to 1 is better) indicates how well-separated the clusters are.
- The **Davies-Bouldin Index** (lower is better) tells you how similar each cluster is to the one that is most similar to it.
- **Conclusion:** Both models did about the same, but K-Means had slightly more compact clusters based on the metrics we saw in the notebook.

4. Ethical and Practical Issues

4.1 Possible Risks

1. **Bias:** The model will make clusters that don't represent all types of wine if the training data is biased (for example, if it only includes wines from one vineyard).
2. **Privacy:** Clustering can lead to sensitive profiling or the "re-identification" of anonymous individuals through behavioral patterns in a consumer context (like the Mall Customer dataset), but this doesn't apply to wine.
3. **Misinterpretation:** The model finds patterns in unsupervised learning, but it doesn't know why they are there. People might misinterpret these clusters, which could lead to bad business choices.

4.2 Mitigation Strategies

- **Anonymization:** Ensure all personally identifiable information (PII) is removed before clustering.
- **Transparency:** Always report the limitations and assumptions made during the modeling process.
- **Human-in-the-Loop:** Have people who know a lot about the subject check to see if the clusters that the algorithm makes make sense in the real world.

5. Key Takeaways

Our group found that PCA is a really important tool for handling data with many different parts. We wouldn't have known if our clustering models made any sense without it. We also saw that Hierarchical clustering gives a much clearer idea of how data points connect to each other, especially with those tree-like dendrograms, even though K-Means gets the job done faster.

6. Repository and Commit Reference

- Repository Link: https://github.com/Muditha-Kumara/Applied-artificial-intelligence/tree/main/2.1
- Commits: 6ba27dc2f41c60120bb10bd25e2209f0788f71e3